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Seminar Thesis

Nowcasting Inflation using Machine Learning Methods

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1 Introduction

Forecasting inflation plays an important role in understanding and managing the economy. It is an essential part for central banks since they rely on accurate inflation forecasts when setting monetary policy. Governments use them for economic planning and the private sector depends on them for decision-making. For this reason, producing accurate inflation forecasts is very important as it enables policymakers to implement efficient and effective policy measures by monetary authority (Narmandakh, 2022). A major challenge in inflation is identifying which variables actually help predict future inflation. Traditional models often struggle because they rely on choosing a small set of variables even though modern datasets contain hundreds of potential predictors. In other words, standard empirical models are limited by the curse of dimensionality which makes them less effective than methods that can handle large datasets and translate them into more useful measures (Kotchoni et al., 2019). Traditional econometric methods such as ARIMA, VAR models, state-space approaches and DSGE frameworks have traditionally been the main tools used for forecasting inflation (Faust & Wright, 2013). Over the past decades a wide range of models including simple univariate time series to fully developed structural macroeconomic models have been developed and used by central banks to forecast key macroeconomic indicators such as inflation (Narmandakh, 2022). As mentioned before, these models come short when it comes to capturing the complex and evolving dynamics of forecasting inflation. Emerging machine learning methods offer a robust and compelling alternative. Techniques such as Ridge, Lasso, Random Forest and Gradient Boosting algorithms are capable of flexibly capturing nonlinear patterns and complex interactions in high-dimensional settings without relying on strict assumptions. An increase in evidence shows that machine learning based approaches often outperform traditional approaches especially in periods of structural change or unusual economic conditions (Moshiri & Cameron (2000); Medeiros et al. (2019); Coulombe et al. (2022) ; Masini et al. (2023)). More recently, advances in computing power and the growing availability of large datasets have increased interest in machine learning methods as potential alternatives to the traditional statistical models employed by monetary authorities. Machine learning techniques offer the ability to better handle main issues such as nonlinearity, multi-collinearity and dimensionality. In addition, these methods allow researchers to more effectively manage the bias-variance trade-off which is very important for building forecasting models leading to more accurate forecasts (Narmandakh, 2022). Since forecasting is very important it is necessary to first assess a models ability to produce accurate predictions and nowcasts. This paper focuses on

nowcasting inflation and evaluating the performance of different models in this setting. In a first step we give an overview of the main ideas and findings from existing literature on machine learning approaches to inflation prediction. After that, we explain the methods used in this paper including their advantages and disadvantages to other methods. We also outline the overall structure of our predicting framework including data sources, data preparation and our variables used. Our analysis is an approach to nowcasting inflation. We apply several machine-learning methods to the same dataset, using a wide range of macroeconomic variables. These models are used to produce out-of-sample inflation nowcasts which we evaluate with common accuracy measures. In the final step, we compare the results of the models to see the best performances.

2 Literature Review

Given the broad relevance of the topic numerous studies in the literature examine the factors that influence inflation. Over the past decade the use of machine learning and big data has expanded rapidly in the literatures relevant to forecasting inflation. The following literature review gives an overview of some important findings from existing studies. Recent papers using machine learning to predict inflation include the study by Medeiros et al. (2019). For example, they applied several ensemble machine learning methods to forecast U.S. inflation and found that the random forest model outperformed all other approaches. Inoue and Kilian (2008) for example examined U.S. inflation forecasts using Lasso and Ridge regression methods. Another paper that adds to this subject is from Maehashi and Shintani (2020). They compare the forecasting performance of traditional factor models with nine different machine-learning methods. They focus on seven key macroeconomic variables for Japan including inflation. Their results show that machine learning methods tend to perform especially well when predicting further into longer forecast horizons. They also find that combining factor models with machine learning methods leads to better forecasts than using either approach on its own. Önder and Akkoç (2021) show that shrinkage-based machine learning methods perform particularly well for variable selection in inflation forecasting. Their results indicate that Lasso and Elastic Net dominate traditional econometric approaches for predicting Turkish inflation. There is also a lot of growing literature on broader macroeconomic forecasting using machine learning. For example, (Coulombe et al., 2022) examines which underlying feature of machine learning methods explain their forecasting gains over standard macroeconomic approaches. Masini et al. (2023) provides a no exhaustive review of recent developments in machine learning and high dimensional statistics for time series modelling and forecasting. Their paper covers both linear and nonlinear approaches. Modugno (2013) proposes a framework to

nowcast and forecast inflation using high frequency data such as weekly and daily indicators instead of relying only on monthly information. Their results show that using high-frequency data improves inflation accuracy compared to models based solely on monthly data. In the paper of Giannone et al. (2008) the authors develop an approach to assess how intra-monthly data releases affect current quarter nowcasts of GDP growth. The method is designed to handle large datasets with staggered release dates. Beck et al. (2023) examine whether high-frequency household scanner data combined with machine learning methods can improve real-time nowcasts of German inflation. The authors nowcast inflation at different levels. Their results show that high-frequency data lead to improvements over time-series models.

3 Data

3.1 Data Source

Table 1: Data overview

Start	End	N_obs	Frequency	Variables
Feb 1994	Aug 2025	380	Monthly	28

Our analysis is based on a monthly macroeconomic dataset constructed from several and widely used data providers. The core of the dataset comes from the Federal Reserve Economic Data (FRED). It is a comprehensive and reliable source of macroeconomic information in the United States. Since it also covers the entire U.S. economy and have consistent methodologies it is very representative for economic conditions and suitable for our analysis. In addition to the FRED series we include the Bloomberg Index and monthly gold prices from financial data provider. We include a wide range of variables that can affect inflation, such as the federal funds rate (FEDFUNDS), retail gasoline price (GASREGW), inflation expectations (EXPINF1YR), gold prices (GOLD), and import and export price indices (IR and IQ), among others. The data spans the period from February 1994 to August 2025 and includes 380 monthly observations which is also illustrated in table 1. This long sample period allows us to capture multiple business cycles and high and low inflation time periods. The high-frequency monthly data is important for the inflation dynamics as it allows us to track short-term movements.

Table 2: Description of all Variables

YearMon	Month-Year timestamp	Monthly	-	-
UNRATE	Unemployment Rate (%)	Monthly	FRED	ΔX
UMCSENT	Consumer Sentiment Index	Monthly	FRED	ΔX
TOTALSL	Total Consumer Credit Outstanding	Monthly	FRED	$\Delta \log(X)$
CES3000000008	Avg Hourly Earnings, Manufacturing	Monthly	FRED	$\Delta \log(X)$
PCE	Personal Consumption Expenditures	Monthly	FRED	$\Delta \log(X)$
RRSFS	Real Retail Sales	Monthly	FRED	$\Delta \log(X)$
FEDFUNDS	Federal Funds Rate (%)	Monthly	FRED	ΔX
M2SL	M2 Money Stock	Monthly	FRED	$\Delta \log(X)$
GS2	2-Year Treasury Yield	Monthly	FRED	ΔX
BOPGSTB	Trade Balance: Goods & Services	Monthly	FRED	ΔX
IR	Import Price Index	Monthly	FRED	$\Delta \log(X)$
IQ	Export Price Index	Monthly	FRED	$\Delta \log(X)$
DCOILWTICO	WTI Crude Oil Price	Daily	FRED	$\Delta \log(X)$
GASREGW	Retail Gasoline Price	Weekly	FRED	$\Delta \log(X)$
INDPRO	Industrial Production Index	Monthly	FRED	$\Delta \log(X)$
PPIACO	PPI: All Commodities	Monthly	FRED	$\Delta \log(X)$
HOUST	Housing Starts	Monthly	FRED	$\Delta \log(X)$
PERMIT	Building Permits	Monthly	FRED	$\Delta \log(X)$
EXPINF1YR	1-Year Inflation Expectations	Monthly	FRED	ΔX
REAINTRATREARATIYE	1-Year Real Interest Rate Expectations	Monthly	FRED	ΔX
CSUSHPinsa	National Home Price Index (Case-Shiller)	Monthly	FRED	$\Delta \log(X)$
CUUR0000SEHA	CPI: Rent of Primary Residence	Monthly	FRED	$\Delta \log(X)$
APU000072610	Average Electricity Price (per kWh)	Monthly	FRED	$\Delta \log(X)$
PCU335311335311	PPI: Electric Power & Transformer Mfg.	Monthly	FRED	$\Delta \log(X)$
STLFSI4	St. Louis Fed Financial Stress Index	Monthly	FRED	X
inflation	CPI Inflation	Monthly	FRED	X
GOLD	Gold Price	Monthly	goldprice.org	$\Delta \log(X)$
BCOM	Bloomberg Commodity Index	Monthly	Bloomberg	$\Delta \log(X)$

Note: ΔX denotes first differences in levels, $\Delta \log(X)$ denotes first differences of the logarithm, and X indicates that the variable enters in levels.

After downloading all data, they were removed of missing values and merged into one data set. All series were aligned by their monthly timestamp (“YearMon”) which ensures that variables measured at different times or from different sources are consistently matched. Table 2 provides an overview of all variables included in our analysis, including their definitions, data frequency and data sources. All variables are transformed where necessary to ensure comparability and stationarity. Figure 4 and Figure 5 in the appendix show the plots for all the variables.

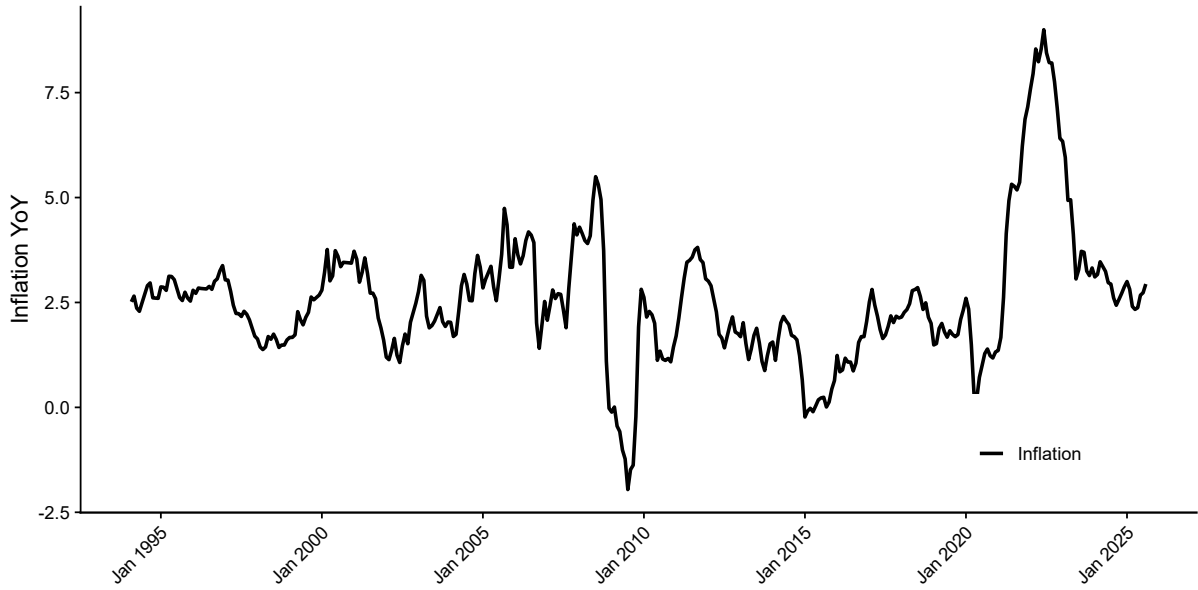


Figure 1: Inflation Plot

Figure 1 shows the inflation series that is used as the dependent variable in our analysis. Inflation is measured as the year-on-year rate meaning that inflation is calculated on an annual basis but observed at a monthly frequency. Many of the real variables are strongly correlated as it is common in macroeconomics data. This relationship is shown in the correlation matrix in the Appendix (Figure 6).

We aim to predict this inflation series using different methods.

3.2 Data Preparation

Before the data preprocessing stage, the daily series DCOILWTICO and the weekly series GASREGW were converted to monthly frequency. Monthly observations were constructed by retaining the last available observation in each calendar month.

Next, the time-series properties of the variables were examined, with a focus on stationarity. After assembling all series from the sources described above, the dataset was prepared for the subsequent prediction exercise. Because the variables cover different economic categories such as prices, interest rates and financial indicators, they are measured in different units and often display pronounced long-run trends. Using such series in levels may give rise to spurious relationships and an unstable estimate. Stationarity was assessed using Augmented Dickey-Fuller (ADF) tests applied to each numerical variable individually. The results show that most series are non-stationary in levels which is also shown in table 6 in the appendix, indicating the presence of a unit root. Only a few variables, such as the trade balance, the financial stress index, and inflation, were stationary without additional transformation. To account for non-

stationarity, most variables were transformed before estimation. Variables expressed in levels were log-transformed where appropriate and then differenced, producing monthly changes that vary more smoothly over time. This transformation removes long-run trends and yields series that fluctuate around a stable mean. Inflation was retained in levels, as it already captures changes over time. To guide model specification, the autocorrelation and partial autocorrelation functions (ACF and PACF) of inflation examined, suggesting that one lagged value of inflation may contain useful predictive information. In addition, a one-period lag of inflation was included as an explanatory variable to capture short-term inflation persistence.

Finally, descriptive statistics were computed for all variables, including the mean, standard deviation, minimum, maximum and median. These statistics summarize the scale and variability of the data over the sample period and are reported in the appendix (table 5).

After completing data preparation, the dataset was divided into training and test samples using a standard fixed hold-out time-series split, with approximately 70% of the observations used for training and the remaining 30% reserved for testing. Consistent with the time-series nature of the data, the split was implemented chronologically rather than randomly. The full sample spans the period from February 1994 to August 2025. The training sample covers February 1994 to mid-2016, while the test sample comprises the remaining observations until August 2025. To ensure comparability across predictors, all explanatory variables were standardized prior to estimation. Standardization was performed using the mean and standard deviation computed from the training sample, and the same transformation was applied to the test sample. This approach avoids look-ahead bias and ensures that all variables enter the model on a common scale.

4 Methodology

4.1 ARIMAX

There is a wide range of models used for prediction and nowcasting. Among these, the ARIMAX model stands out for its ability to incorporate external information which can lead to more robust and accurate results. ARIMAX stands for Autoregressive Integrated Moving Average with exogenous variables and is an extension of the standard ARIMA framework. While ARIMA models rely mostly on the dynamics of a time series, ARIMAX models additionally include exogenous variables, representing external factors that may influence the target series (*What Is an ARIMAX Model?*, n.d.). We applied this model because inflation dynamics are driven not only by their past behaviour but also by a wide range of variables. This

model allows us to incorporate these variables as exogenous regressors while still modelling autocorrelation and trends in inflation. This is important in our case because many of the explanatory variables, after being transformed to ensure stationarity by using log differences, include current and forward-looking information that helps explain inflation beyond what can be captured by its own past values alone.

4.2 LASSO

Another important machine learning method used in our analysis is LASSO regression. It stands for Least Absolute Shrinkage and Selection Operator and is a linear regression technique that performs both variable selection and regularization. Its main purpose is to reduce overfitting by shrinking the coefficients of less important variables toward zero and in some cases setting them exactly to zero. By adding a penalty term to the standard regression model, LASSO helps identify which variables are truly relevant for explaining the dependent variable. This is useful in a real-world dataset where many explanatory variables are highly correlated with each other, a problem known as multicollinearity. In such cases LASSO is able to select the most informative predictors while excluding the redundant ones (*What Is Lasso Regression?*, n.d.). We use LASSO regression because our dataset includes a large number of variables that may affect inflation while not all of them are equally important. In such settings, traditional regression models can easily overfit the data or suffer from multicollinearity due to strong correlations between the explanatory variables. As mentioned before, LASSO helps address these issues by shrinking the coefficients of less relevant variables toward zero and effectively removes them from the model. This allows the model to automatically select the most informative predictors and ignores the weak ones. This leads to a simpler and more stable model which is easier to interpret.

4.3 Random Forest

A random forest is a machine learning model that combines many decision trees to make more accurate and stable predictions. Instead of relying on a single tree, a random forest builds multiple trees, each slightly different from the other and then averages their results. This is created by using a technique called bagging. With bagging each decision tree is trained on a randomly selected subset of the training data. Although each subset contains the same number of observations as the original dataset some observations may appear multiple times while others may be left out. Because each tree sees a different version of the data, the trees make different errors. By combining the predictions of all trees, random forests reduce overfitting and usually perform better than a single decision tree (*Random Forests*, n.d.). We apply

Random Forest because traditional linear models may miss these nonlinear relationships and interactions between variables. Random Forest can automatically capture such patterns without requiring us to specify them in advance.

4.4 XGBoost

XGBoost short for extreme Gradient Boosting is a more advanced machine learning method that is designed to achieve higher accuracy while remaining efficient. The core idea behind gradient boosting is to build a strong model by combining many weak models usually simple decision trees. Each new tree is trained to focus on the mistakes made by the previous trees gradually improving the overall prediction performance. This works by minimizing prediction errors step by step using gradient descent. At each step the model fits a new tree to the residuals of the previous model and the final prediction is obtained by combining the predictions of all trees (*What Is XGBoost?*, n.d.). We also apply XGBoost to test whether allowing the model to learn if this method leads to a better result. Since XGBoost adapts flexibly to the data by correcting its own mistakes, it is suitable for our analysis.

4.5 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction method that aims to simplify a dataset by reducing the number of explanatory variables while preserving as much of the original information as possible. It transforms a set of correlated variables into a smaller number of uncorrelated components known as principal components. PCA works by identifying directions in the data that capture the greatest variation. These directions are determined using eigenvalues and eigenvectors of the covariance matrix. The components associated with the largest eigenvalues are retained and used to represent the data in a lower dimensional space resulting in a simpler and more compact dataset (*Principal Component Analysis (PCA)*, n.d.).

5 Results

In the following section we will discuss the results of each model and compare their performance. We then focus more detail on LASSO and XGBoost as these models perform best relative to the others considered in our analysis.

Table 3: Model Evaluation Metrics

Model	RMSE	MAE	SMAPE
LASSO	0,2963	0,2103	0,0948
Random Forest	1,7959	0,9961	0,2748
XGBoost	1,4662	0,7579	0,1938
PCA Regression	2,2707	1,642	0,4556
ARIMAX	2,1788	1,3573	0,4066

To evaluate the performance of the models we rely on three common accuracy measures. Table 3 shows the values for RMSE, MAE and SMAPE for each model. The Root Mean Squared Error (RMSE) measures the average magnitude of prediction errors, placing a larger penalty on bigger deviations. The Mean Absolute Error (MAE) measures the average absolute difference between predicted and observed values and is directly interpretable in the units of the data. The Symmetric Percentage Error (SMAPE) expresses errors in relative terms and allows for comparisons across models. The results show that the LASSO model outperforms all other approaches, achieving the lowest values for RMSE, MAE, and SMAPE. This indicates that a sparse linear specification is particularly effective in capturing the predictive relationship in the data. Considering the remaining machine learning models, XGBoost performs better than both Random Forest and PCA but accuracy still lower than LASSO across all metrics. This means that while nonlinear effects may be present, they do not provide sufficient additional explanatory power. As a traditional method, ARIMAX performs relatively poor to the machine learning methods, and this may be related to its reliance on linear relationships and strong model assumptions which may limit its ability to capture more complex patterns. The predictions for Random Forest, ARIMAX and PCA are included in the appendix. Figure 10 in the appendix presents the scree plot from the principal component analysis, showing the share of variance explained by each component. Figure 11 in the appendix displays the scatter plot of predicted and actual inflation for the PCR model, illustrating the model's fit to the observed data.

5.1 LASSO



Figure 2: LASSO: Actual vs Predicted

Figure 2 shows the actual inflation rate and the values predicted by the LASSO model from Feb 1994 to Aug 2025. The solid grey line represents the observed inflation while the green line shows the models predictions. The vertical dashed red line marks the split between the in-sample period which is used for model estimation and the out-sample period which is used for evaluation. The predicted series closely follows the actual inflation path both before and after the sample split. The model captures the main movements and turning points in inflation including periods of moderate variation as well as sudden changes. While some short-term spikes are slightly smoothed the general level and direction of inflation are well replicated. The model maintains a strong fit in the out-of-sample period indicating stable performance rather than overfitting to the training data.

Table 4: Lasso Model Coefficients

Variables	Coefficient
inflation_lag1	1,08
PPIACO	0,16
IR	0,05
GASREGW	0,04
CUUR0000SEHA	0,04
PCU335311335311	0,03
PCE	0,02
UMCSENT	-0,02
BOPGSTB	-0,01
STLFSI4	-0.002
CES3000000008	-0.001

Table 4 shows the coefficients selected by the LASSO model. The lagged inflation variable has by far the largest coefficient showing that past inflation is the strongest driver of current inflation. Inflation expectations also play an important role but their effect is much smaller compared to the lagged inflation term. Price and cost pressures also effect inflation but with much smaller weights. While a 1 standard deviation increase in last month's inflation is associated with a 1.08 standard deviation increase in current inflation, increases in producer prices (PPIACO), import prices (IR), retail gasoline prices (GASREGW), rents (CUUR0000SEHA), electricity related producer prices (PCU335311335311), and personal consumption expenditures (PCE) are associated with positive but considerably smaller increases in inflation with coefficients ranging between 0.02 and 0.16 standard deviations. In contrast, consumer sentiment (UMCSENT), the trade balance (BOPGSTB), financial stress (STLFSI4), and wages (CES3000000008) enter with negative coefficients. Rest of the variables are shrunk to zero. The actual and predicted LASSO values can be seen in table 7 in the appendix.

LASSO effectively selects only the most informative predictors by shrinking coefficients and setting less relevant ones to zero. The strong performance can be explained by the fact that the most informative predictor of inflation is its own lag. Inflation tends to be highly persistent meaning that past inflation values already contain a large amount of information about current inflation. Once these lagged inflation terms are included in the model much of the variation in inflation can be explained through the linear relationship. LASSO effectively recognizes this

by selecting the lagged inflation variable and shrinking the coefficients of less informative variables towards zero. Because the dominant predictor is linear the overall problem becomes largely linear than nonlinear. This might explain why LASSO outperforms nonlinear models in this case.

5.2 XGBoost

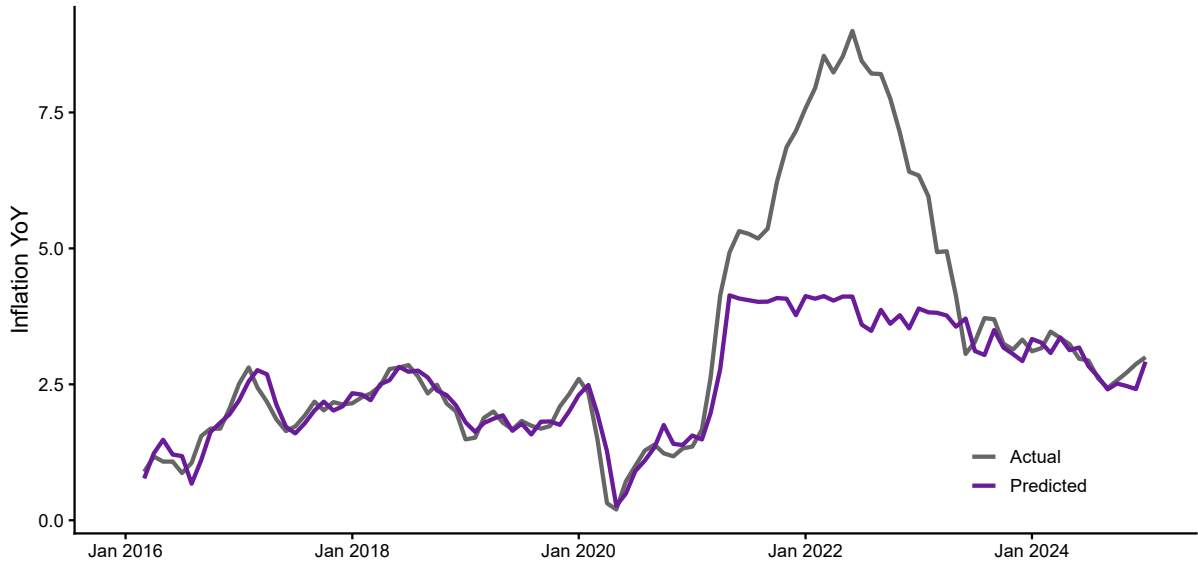


Figure 3: XGBoost: Actual vs Predicted

Figure 3 shows the inflation rate and the values predicted by the XGBoost model for the years Jan 2016 to Aug 2025. Here again the grey line represents observed inflation while the purple line shows the models predictions. The predicted series follows the general movement of inflation closely especially during periods of gradual changes and stabilization. XGBoost captures the broad upward shift in inflation starting in 2020 quite well. The feature importance plot indicates that inflation receives substantially greater weight in the XGboost model than all other variables (Figure 10). The timing and the direction of these movements are largely aligned with the observed data indicating that the model is able to learn important nonlinear relationships between inflation and the predictors. However, the model fails to fully capture the sharp peak observed during the inflation surge. While actual inflation rises steeply XGBoost produces a smoother and clearly lower response underestimating the magnitude of the spike. The strong performance of XGBoost can be explained by its boosting structure which allows the model to learn complex and nonlinear patterns in the data. By fitting trees to the errors of previous models, XGBoost improves its predictions and focuses on variables that are most informative at stage. At the same time, XGBoost is unable to fully capture the sharp inflation

spike is that COVID 19 represents an exceptional shock. During this time, inflation was driven by many different events. In the years before the pandemic inflation was relatively stable and did not have movements of this comparable magnitude. Since XGBoost learns from past relationships in the data it may struggle when the underlying economic structure changes abruptly. When an event occurs such as COVID 19 the model tends to smooth the response rather than reproducing the full extent of the spike. Moreover, the COVID 19 years are often treated as a special case in empirical research which is why many studies exclude them or separately handle them.

6 Shortcomings

The key idea of this paper is that it focuses on nowcasting rather than forecasting. Forecasting inflation at longer horizons is more challenging and would go beyond the scope of this study. If this paper had focused on forecasting rather than nowcasting several additional steps would have been required. Forecasting inflation involves predicting future values several periods ahead using only information available at the time of prediction. This would require shifting the target variable forward and restricting all explanatory variables to lagged values which reduces the amount of usable information a lot. In that setting the strong influence of recent inflation becomes weaker as the forecast horizon increases. This makes predictions more uncertain and increases the importance of capturing relationships between variables. To deal with this more advanced machine learning methods are often needed. Also, forecasting requires a more careful evaluation strategy. Instead of using a single train-test split, rolling or expanding window approaches are commonly applied. These methods re-estimate the model over time and test it on future periods which helps assess how stable the model performance is.

Nowcasting is still highly important because it is important to test models and for economic research.

7 Conclusion

This paper analyses how different models can be used to nowcast inflation. We compared several commonly used methods, including ARIMAX, LASSO, Random Forest, XGBoost and PCA. These approaches differ in their complexity and assumptions. All models are evaluated using the same dataset and error measures to ensure a good comparison. The results show clear differences in performance across the models. In our case LASSO achieves the best results. It is able to track inflation closely and produces the most accurate results overall. XGBoost also

performs well and clearly improves upon Random Forest and PCA although it has difficulties capturing large and sudden inflation spikes, especially during exceptional periods such as the COVID 19 pandemic. Random Forest and ARIMAX show weaker performance and PCA model performs worst suggesting that important information is lost when the data are strongly reduced.

The findings show that simple models can be very effective for inflation nowcasting especially when inflation is strongly driven by its past values.

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9 Appendix

Figures

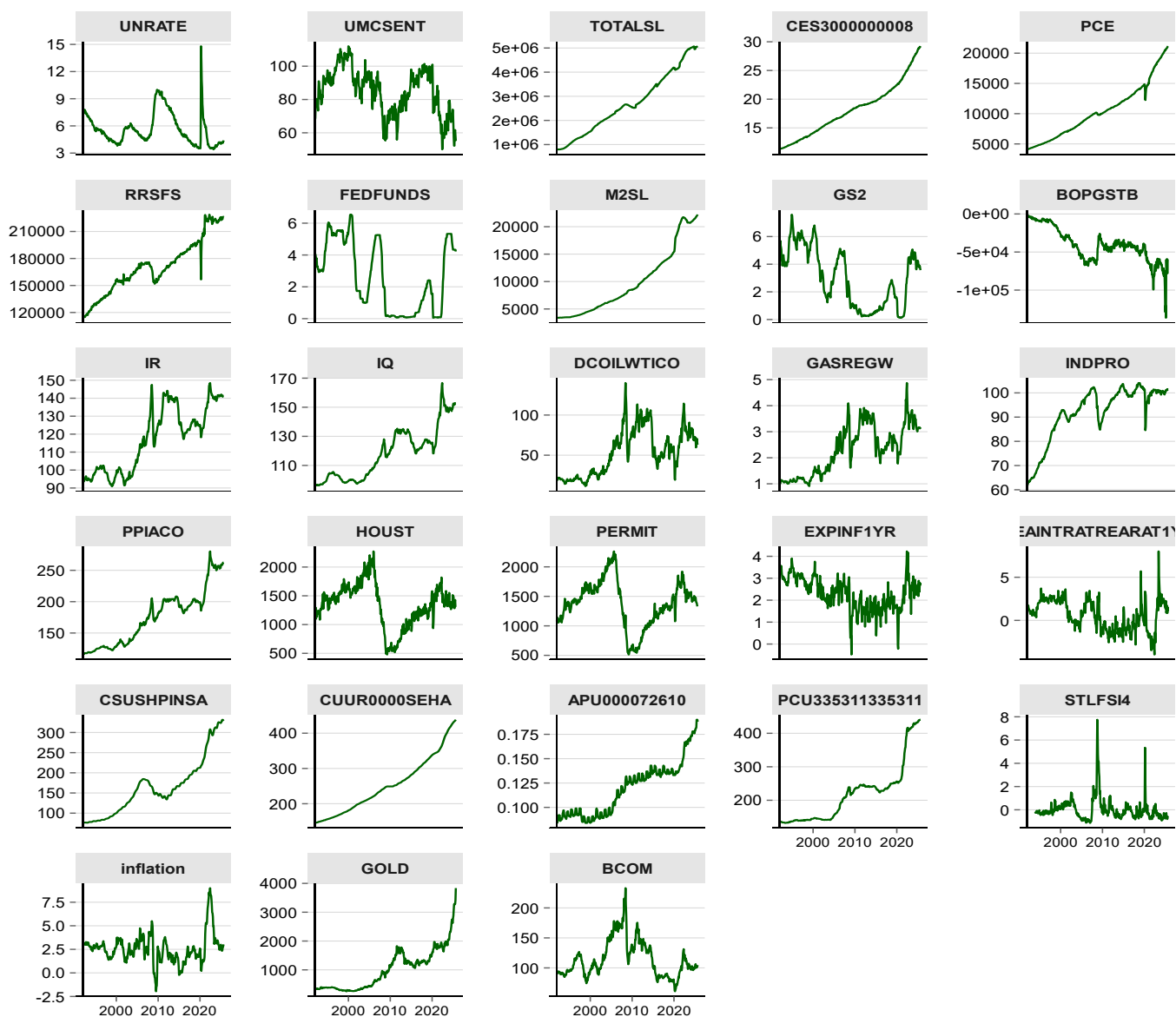


Figure 4: All Time Series Plotted

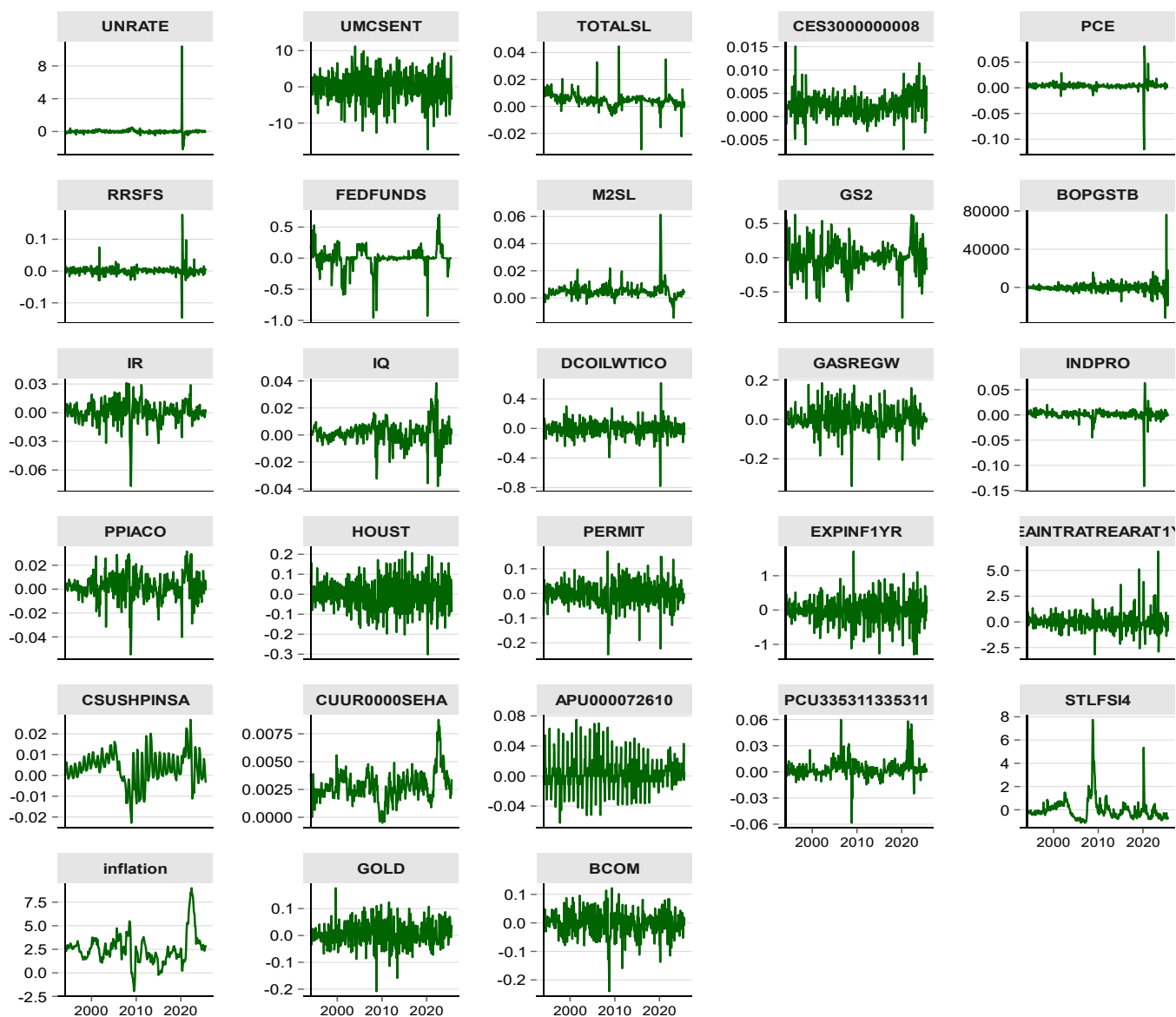


Figure 5: All Time Series: Differenced

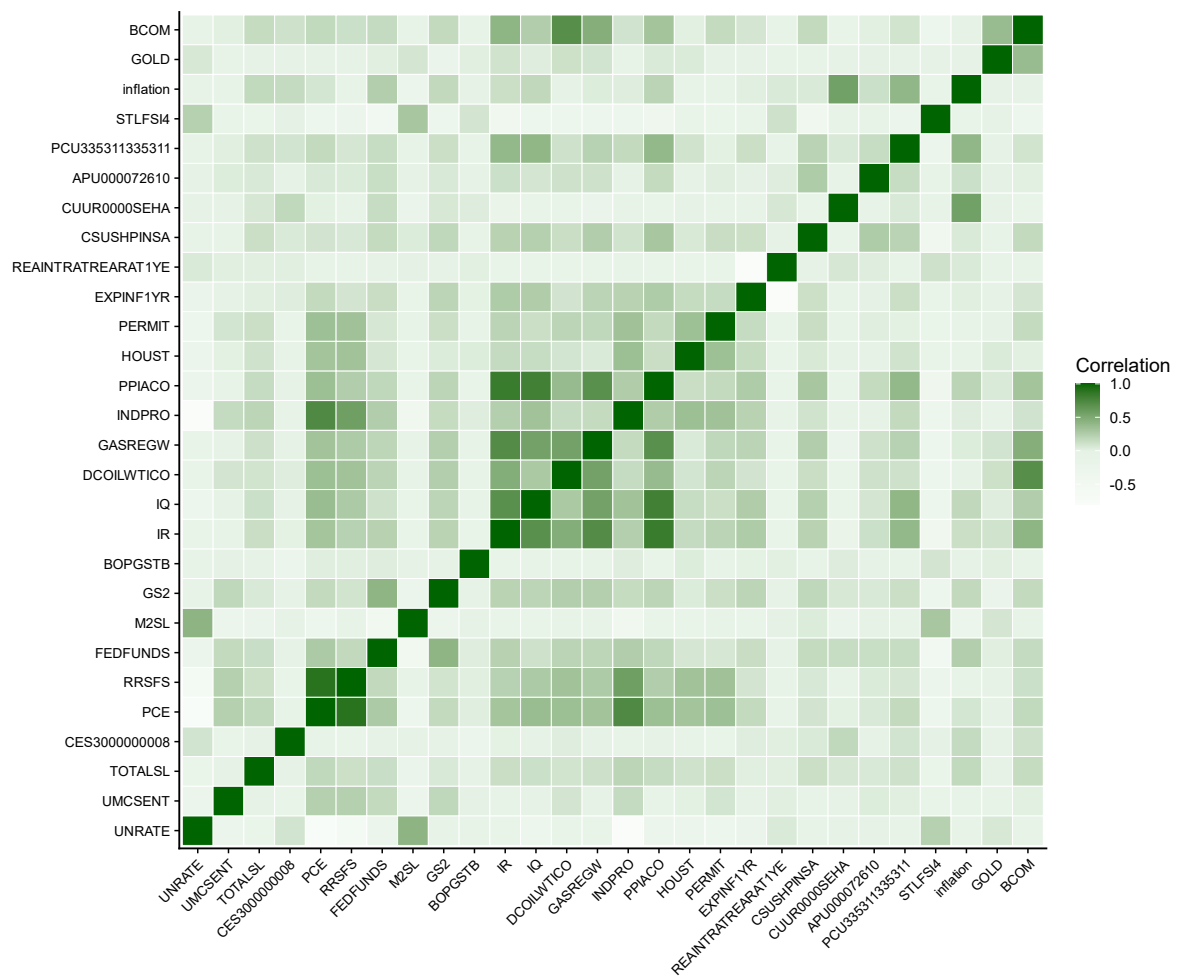


Figure 6: Correlation Matrix of Variables

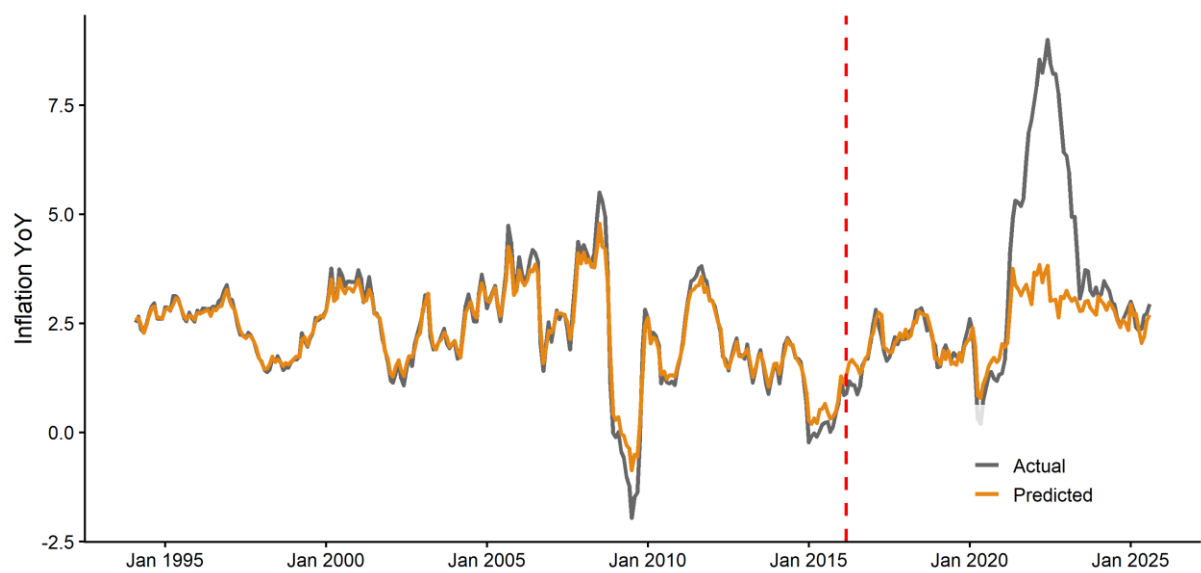


Figure 7: RANDOM FOREST: Actual vs Predicted

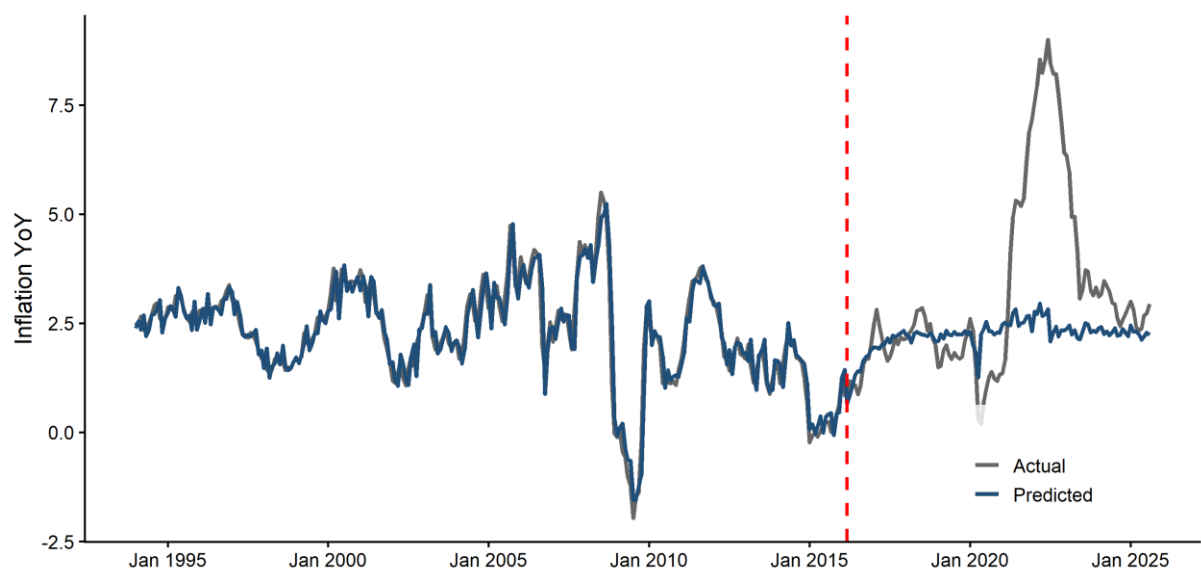


Figure 8: ARIMAX: Actual vs Predicted

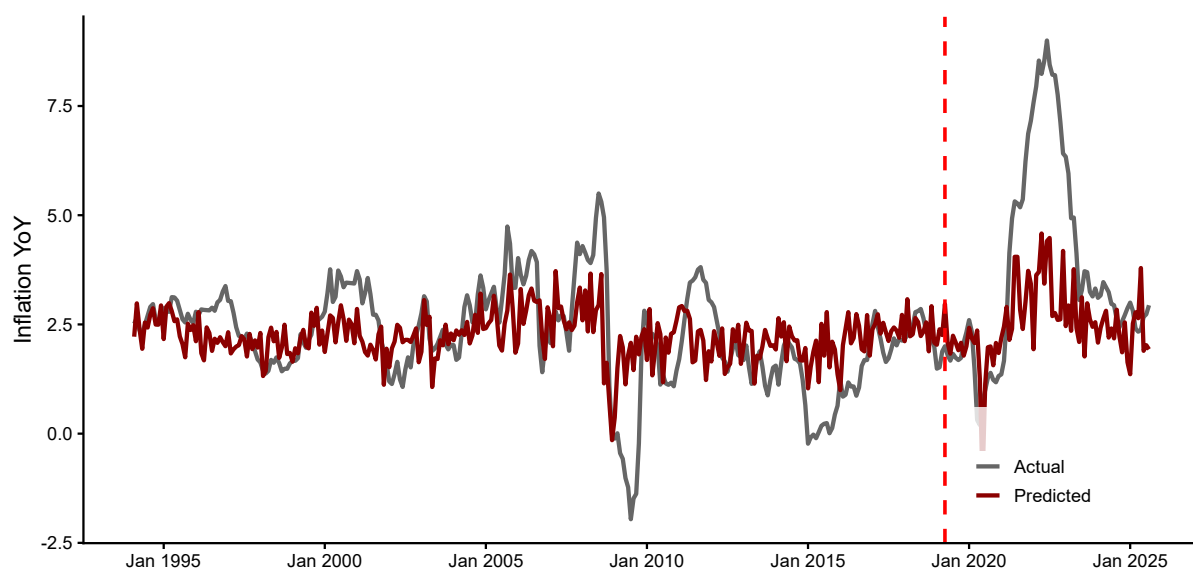


Figure 9: PCA: Actual vs Predicted

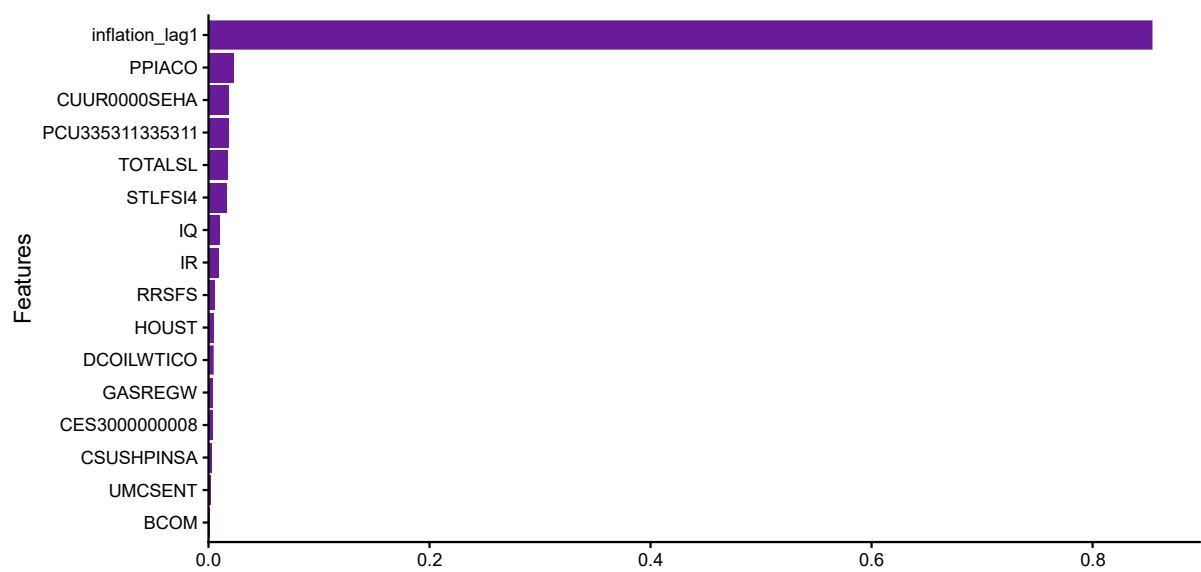


Figure 10: XGBOOST: Feature Importance Plot

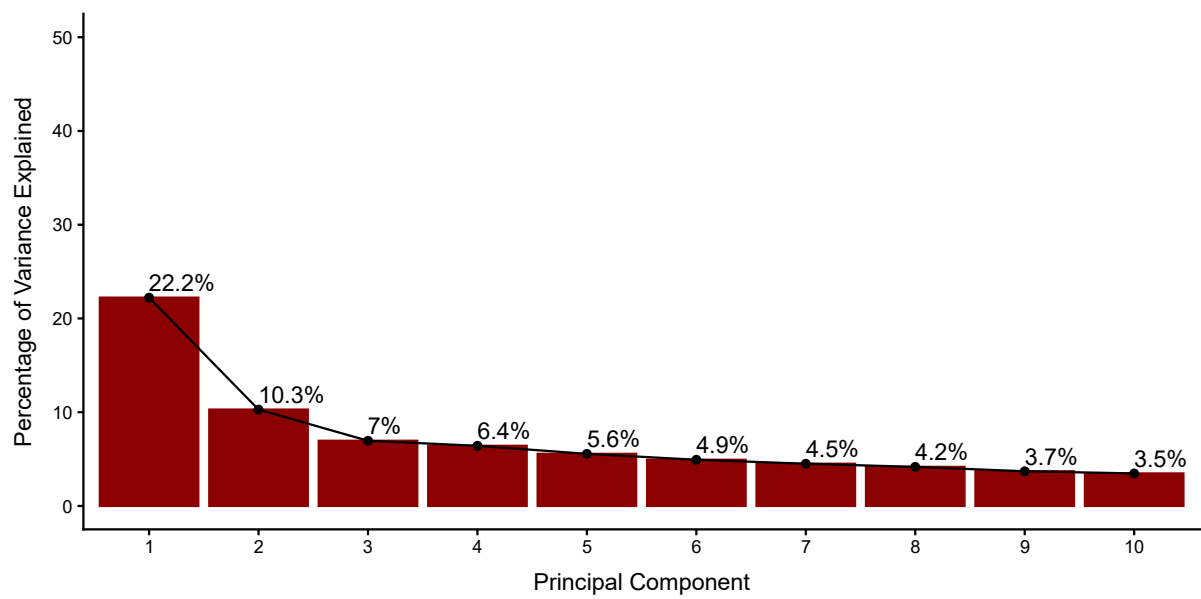


Figure 11: PCA: Share of Variance

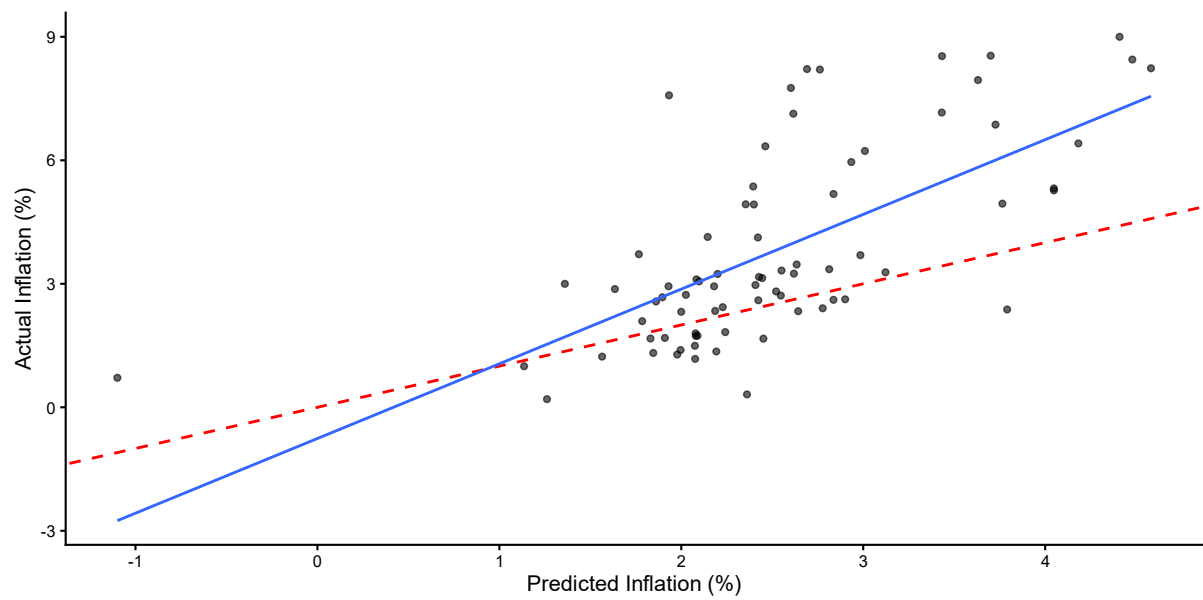


Figure 12: PCR: Actual vs Prediction

Tables

Table 5: Descriptive Statistics of Variables

Variable	mean	sd	min	max	median
UNRATE	5,65	1,782	3,4	14,8	5,2
UMCSENT	84,976	13,894	50	112	88,1
TOTALSL	2692757,43	1261025,93	797276,31	5074930,91	2566995,36
CES3000000008	18,111	4,559	11,24	29,23	17,95
PCE	10385,281	4454,37	4084,7	21152,2	9976,7
RRSFS	170584,38	29999,448	114052	228772	169138
FEDFUNDS	2,634	2,175	0,05	6,54	2,28
M2SL	9800,057	5887,11	3381,2	22212,5	8027
GS2	2,998	2,093	0,12	7,59	2,86
BOPGSTB	-40671,599	22391,201	-136419	-831	-41548,5
IR	117,171	17,953	91	148,5	120
IQ	117,992	17,653	95,7	166,7	118,2
DCOILWTICO	53,307	29,141	11,37	139,96	51,67
GASREGW	2,285	0,943	0,907	4,872	2,299
INDPRO	91,927	11,205	61,462	104,1	95,955
PPIACO	174,398	43,877	115,6	280,251	177,4
HOUST	1338,822	384,491	478	2273	1375,5
PERMIT	1374,203	395,992	513	2263	1404,5
EXPINF1YR	2,291	0,74	-0,481	4,227	2,334
REAINTRATREARAT1YE	0,628	1,762	-3,966	8,006	0,736
CSUSHPINS	162,833	69,942	75,658	331,671	152,339
CUUR0000SEHA	251,946	78,763	145,4	438,212	246,681
APU000072610	0,119	0,027	0,084	0,19	0,124
PCU335311335311	217,823	86,241	132,2	440,55	223,1
STLFSI4	0	0,981	-1,134	7,748	-0,213
inflation	2,576	1,54	-1,959	8,999	2,539
GOLD	997,647	699,771	254,6	3836,234	918,5
BCOM	114,176	30,879	60,898	233,034	104,683

Table 6: Results of Augmented Dickey-Fuller Test

Variable	ADF_stat	p_value	Decision
CES3000000008	0,04189707	0,99	Non-stationary
HOUST	-1,31386356	0,8667876	Non-stationary
PERMIT	-1,58478679	0,75240161	Non-stationary
PCE	-1,68730404	0,70911799	Non-stationary
CUUR0000SEHA	-1,7272196	0,69226532	Non-stationary
APU000072610	-1,84312484	0,64332918	Non-stationary
GOLD	-1,84552641	0,64231522	Non-stationary
CSUSHPinsa	-1,85133385	0,63986327	Non-stationary
PCU335311335311	-1,9920851	0,58043695	Non-stationary
GS2	-2,0675083	0,54859265	Non-stationary
UMCSENT	-2,08590685	0,54082463	Non-stationary
GASREGW	-2,09964429	0,53502458	Non-stationary
DCOILWTICO	-2,11251183	0,5295918	Non-stationary
RRSFS	-2,19629503	0,49421785	Non-stationary
IR	-2,28488852	0,45681295	Non-stationary
BCOM	-2,29832716	0,45113905	Non-stationary
UNRATE	-2,38736695	0,41354573	Non-stationary
EXPINF1YR	-2,61079619	0,31921208	Non-stationary
IQ	-2,61338076	0,31812085	Non-stationary
PPIACO	-2,71141733	0,27672901	Non-stationary
REAINTRATREARAT1YE	-2,73139748	0,26829323	Non-stationary
FEDFUNDS	-2,87625512	0,20713315	Non-stationary
INDPRO	-3,02268147	0,14531076	Non-stationary
TOTALSL	-3,08177993	0,12035891	Non-stationary
M2SL	-3,24266825	0,08089073	Non-stationary
BOPGSTB	-7,11212334	0,01	Stationary
STLFSI4	-4,21207352	0,01	Stationary
inflation	-4,13956226	0,01	Stationary

Table 7: LASSO: Actual vs Predicted Values

Date	actual	pred	error	abs_error	sq_error
Mar 2016	0,892	1,059	0,167	0,167	0,028
Apr 2016	1,173	1,152	-0,021	0,021	0,0004
May 2016	1,078	1,468	0,389	0,389	0,1516
Jun 2016	1,079	1,396	0,317	0,317	0,1004
Jul 2016	0,868	1,148	0,279	0,279	0,0781
Aug 2016	1,055	0,899	-0,157	0,157	0,0246
Sept 2016	1,549	1,162	-0,387	0,387	0,1498
Oct 2016	1,686	1,675	-0,011	0,011	0,0001
Nov 2016	1,684	1,676	-0,008	0,008	0,0001
Dec 2016	2,051	1,954	-0,097	0,097	0,0094
Jan 2017	2,51	2,272	-0,238	0,238	0,0567
Feb 2017	2,81	2,53	-0,28	0,28	0,0785
Mar 2017	2,441	2,713	0,272	0,272	0,0738
Apr 2017	2,176	2,562	0,386	0,386	0,1489
May 2017	1,856	2,117	0,26	0,26	0,0677
Jun 2017	1,641	1,887	0,247	0,247	0,0608
Jul 2017	1,725	1,672	-0,053	0,053	0,0028
Aug 2017	1,928	1,802	-0,126	0,126	0,016
Sept 2017	2,181	2,113	-0,068	0,068	0,0046
Oct 2017	2,021	2,196	0,175	0,175	0,0306
Nov 2017	2,172	2,219	0,046	0,046	0,0021
Dec 2017	2,13	2,253	0,123	0,123	0,0152
Jan 2018	2,151	2,303	0,152	0,152	0,023
Feb 2018	2,263	2,21	-0,054	0,054	0,0029
Mar 2018	2,331	2,189	-0,142	0,142	0,0201
Apr 2018	2,471	2,485	0,014	0,014	0,0002
May 2018	2,782	2,662	-0,12	0,12	0,0144
Jun 2018	2,808	2,778	-0,03	0,03	0,0009
Jul 2018	2,854	2,821	-0,033	0,033	0,0011
Aug 2018	2,643	2,741	0,098	0,098	0,0096
Sept 2018	2,332	2,557	0,225	0,225	0,0507
Oct 2018	2,492	2,42	-0,072	0,072	0,0051
Nov 2018	2,147	2,21	0,063	0,063	0,0039
Dec 2018	2,002	1,866	-0,137	0,137	0,0186
Jan 2019	1,488	1,853	0,365	0,365	0,1334
Feb 2019	1,519	1,562	0,043	0,043	0,0018
Mar 2019	1,883	1,808	-0,075	0,075	0,0057
Apr 2019	2,001	2,069	0,069	0,069	0,0047
May 2019	1,796	1,936	0,14	0,14	0,0197
Jun 2019	1,671	1,665	-0,006	0,006	0
Jul 2019	1,826	1,766	-0,06	0,06	0,0036
Aug 2019	1,738	1,727	-0,011	0,011	0,0001

Sept 2019	1,685	1,702	0,017	0,017	0,0003
Oct 2019	1,734	1,649	-0,085	0,085	0,0072
Nov 2019	2,092	1,823	-0,27	0,27	0,0728
Dec 2019	2,32	2,126	-0,193	0,193	0,0373
Jan 2020	2,6	2,313	-0,287	0,287	0,0821
Feb 2020	2,341	2,245	-0,096	0,096	0,0093
Mar 2020	1,494	1,453	-0,041	0,041	0,0017
Apr 2020	0,313	0,149	-0,164	0,164	0,0268
May 2020	0,198	1,19	0,991	0,991	0,9829
Jun 2020	0,717	0,875	0,158	0,158	0,0251
Jul 2020	0,997	1,175	0,178	0,178	0,0317
Aug 2020	1,281	1,239	-0,042	0,042	0,0017
Sept 2020	1,391	1,382	-0,009	0,009	0,0001
Oct 2020	1,23	1,504	0,274	0,274	0,075
Nov 2020	1,176	1,401	0,225	0,225	0,0506
Dec 2020	1,32	1,424	0,104	0,104	0,0108
Jan 2021	1,355	1,846	0,49	0,49	0,2403
Feb 2021	1,668	1,848	0,181	0,181	0,0327
Mar 2021	2,624	2,333	-0,29	0,29	0,0844
Apr 2021	4,137	2,842	-1,295	1,295	1,6768
May 2021	4,926	4,62	-0,306	0,306	0,0939
Jun 2021	5,317	5,202	-0,116	0,116	0,0134
Jul 2021	5,269	5,218	-0,051	0,051	0,0026
Aug 2021	5,181	5,204	0,022	0,022	0,0005
Sept 2021	5,364	5,217	-0,147	0,147	0,0216
Oct 2021	6,227	5,604	-0,623	0,623	0,388
Nov 2021	6,866	6,365	-0,5	0,5	0,2503
Dec 2021	7,159	6,348	-0,812	0,812	0,6588
Jan 2022	7,578	7,285	-0,293	0,293	0,0857
Feb 2022	7,949	7,673	-0,277	0,277	0,0765
Mar 2022	8,541	8,382	-0,159	0,159	0,0251
Apr 2022	8,235	8,327	0,092	0,092	0,0085
May 2022	8,53	8,352	-0,179	0,179	0,0319
Jun 2022	8,999	8,685	-0,314	0,314	0,0986
Jul 2022	8,448	7,928	-0,52	0,52	0,2701
Aug 2022	8,216	7,745	-0,471	0,471	0,2223
Sept 2022	8,206	7,699	-0,507	0,507	0,2572
Oct 2022	7,757	7,592	-0,166	0,166	0,0275
Nov 2022	7,131	7,198	0,066	0,066	0,0044
Dec 2022	6,411	6,478	0,067	0,067	0,0045
Jan 2023	6,34	6,373	0,033	0,033	0,0011
Feb 2023	5,958	5,95	-0,008	0,008	0,0001
Mar 2023	4,931	5,528	0,597	0,597	0,3559
Apr 2023	4,947	4,844	-0,103	0,103	0,0106
May 2023	4,125	4,518	0,392	0,392	0,154

Jun 2023	3,059	3,961	0,902	0,902	0,8139
Jul 2023	3,28	3,004	-0,276	0,276	0,0763
Aug 2023	3,719	3,489	-0,23	0,23	0,0527
Sept 2023	3,695	3,758	0,063	0,063	0,004
Oct 2023	3,247	3,377	0,131	0,131	0,0171
Nov 2023	3,14	3,035	-0,105	0,105	0,011
Dec 2023	3,322	2,861	-0,461	0,461	0,2125
Jan 2024	3,108	3,292	0,184	0,184	0,0337
Feb 2024	3,166	3,316	0,15	0,15	0,0224
Mar 2024	3,469	3,179	-0,29	0,29	0,0841
Apr 2024	3,354	3,508	0,154	0,154	0,0237
May 2024	3,239	3,192	-0,047	0,047	0,0022
Jun 2024	2,97	3,153	0,183	0,183	0,0335
Jul 2024	2,938	3,062	0,123	0,123	0,0152
Aug 2024	2,611	2,693	0,082	0,082	0,0067
Sept 2024	2,433	2,441	0,008	0,008	0,0001
Oct 2024	2,571	2,418	-0,154	0,154	0,0236
Nov 2024	2,714	2,545	-0,17	0,17	0,0288
Dec 2024	2,872	2,776	-0,096	0,096	0,0093
Jan 2025	2,999	3,177	0,178	0,178	0,0316
Feb 2025	2,814	3,026	0,211	0,211	0,0447
Mar 2025	2,406	2,844	0,438	0,438	0,1922
Apr 2025	2,334	2,043	-0,291	0,291	0,0847
May 2025	2,376	2,296	-0,08	0,08	0,0064
Jun 2025	2,673	2,315	-0,357	0,357	0,1277
Jul 2025	2,732	2,793	0,061	0,061	0,0037
Aug 2025	2,939	2,619	-0,321	0,321	0,1028